

The unified compactified picture and the computed 3-vs-6 bridge

(T^N, G) for FFGFT and HLV: why $6 = \mathbf{3} \oplus \mathbf{3}'$, and why the shared $3+1$ is not a shared structure

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Abstract

FFGFT and HLV both reduce to the observable 3+1. This raises a sharp question: if the four circles of FFGFT and the six (plus time) of HLV both end in $\mathbb{R}^3 \times \mathbb{R}_t$, are the 3 and the 6 merely two coordinate descriptions of the same thing – i.e. only a mathematical transformation? This document gives the compact-view answer in one notation and checks it by computation. In the compact picture both models are of the same form: a torus $T^N = (S^1)^N$ carrying a finite symmetry group G , with a Type-I decompactification (the measurement) selecting one circle as time. FFGFT is $(T^4, \mathbb{Z}_3)$; HLV is $(T^7 = T^6 \times S^1, A_5)$. Three reproducible results settle the question. (1) “6 is the doubling of the 3” is literal: the icosahedral 6D representation splits as $\mathbf{3} \oplus \mathbf{3}'$, the two 3D irreducibles of A_5 , differing exactly by $\varphi \leftrightarrow 1 - \varphi$; the 6D character is an integer while each 3D block is irrational (φ). (2) The triangle is the more fundamental, minimal carrier: 3-fold symmetry is crystallographic and lives in 3D, while 5-fold is non-crystallographic and *requires* the 6D embedding. (3) The two models are *reconcilable by containment*: the icosahedral group A_5 contains the 3-fold C_3 (both are symmetries of the same icosahedron), and $T^7 = T^4 \times T^3$, so HLV is FFGFT’s 4D extended by the three perp dimensions ($7 = 4 + 3$, $\text{HLV} \supset \text{FFGFT}$). The protected values $\sqrt{2}$ (Koide 2/3) and $2/\sqrt{5}$ (φ) belong to two *different protected geometries* – FFGFT’s independent 3-fold and HLV’s 5-fold; the groups coexist ($C_3 < A_5$), so this is no incompatibility, and the cut-and-project family $r(\tau) \leq 1$ failing to map one onto the other shows only that interconversion is the wrong question. (Per Doc. 283, $\sqrt{2}$ is *not* inherited from HLV’s icosahedrally protected directions, which give only $2/\sqrt{5}$; it is a separate 3-fold protected choice.) The genuine difference is *which symmetry sector is physically realised*: FFGFT realises the minimal crystallographic 3-fold core (Koide $\theta = 2/9$, empirically confirmed), while HLV adds the 5-fold/perp structure as a testable extension. Hence “only a mathematical transformation” is too weak (the realised sector is a genuine, empirically decidable feature) and “inequivalent structures” is too strong (they are reconcilable, $\text{HLV} \supset \text{FFGFT}$): the precise statement is a shared 3+1 core with HLV the icosahedral extension of FFGFT’s minimal 3-fold core.

.1 Occasion: both reduce to 3+1 – is the rest a transformation?

Doc. 270 separates the reductions of FFGFT ($T^4 \rightarrow 3+1$) and Doc. 283 the FFGFT $\leftrightarrow$ HLV relation (a $\mathbb{Z}_3$ -circulant bridge along a C_3 axis, forked at the symmetry order, $\sqrt{2}$ vs $2/\sqrt{5}$). HLV builds space as a 6D $\rightarrow$ 3D icosahedral cut-and-project and carries time separately (spiral-time). Both models therefore end in the same observable $\mathbb{R}^3 \times \mathbb{R}_t$. The natural question is whether the difference between FFGFT’s 3 and HLV’s 6 is then merely a choice of coordinates – a lossless mathematical transformation of one and the same 3+1. The answer requires a common notation and a computation; both are given here. All numerical statements are reproducible with three short scripts (§.21).

.2 The unified compactified notation (T^N, G)

In the compact representation there is no “space” versus “time” and no “internal” versus “external”: a circle is a circle. Both models are then of one form:

$$\mathcal{M} = T^N = (S^1)^N, \quad G \subset O(N) \text{ acting}, \quad (1)$$

with a Type-I decompactification (the measurement) selecting one circle as time. The remaining circles stay compact (discrete spectrum) or project (space, Type II, Doc. 270). The two models are two instances of (T^N, G) :

	FFGFT	HLV
compact form	$T^4 = (S^1)^4$	$T^7 = T^6 \times S^1$
symmetry group G	$\mathbb{Z}_3$ (3-fold, crystallo-graphic)	A_5 icosahedral (5-fold, non-crystallographic)
space (compact)	3 circles	6 circles = $\mathbf{3} \oplus \mathbf{3}'$
time	one circle (mass circle, $\tilde{T} \cdot m = 1$); scale-linked by the duality, <i>not</i> a split-off factor	one circle (spiral-time); an A_5 -singlet, so the torus <i>factorises</i> $T^6 \times S^1$
how G acts on time	woven in through $\tilde{T} \cdot m = 1$	trivially (singlet)
protected invariant	$r = \sqrt{2}$, Koide $Q = 2/3$, $\theta = 2/9$	$r = 2/\sqrt{5}$, φ
why this N_{space}	3 suffices (3-fold crystallo-graphic)	6 needed (5-fold non-crystallographic)

The whole comparison reduces to one question – *how does G act on the circles of the torus?* The labels “internal/external” are decompactification language and have no place in the compact picture; the correct compact-view distinction for the time circle is group-theoretic: in HLV it is an A_5 -singlet (the torus factorises $T^6 \times S^1$, time decoupled from the spatial symmetry); in FFGFT it is tied to the other circles by the duality $\tilde{T} \cdot m = 1$ (not a direct-product factor). Which circle becomes the observed time is fixed by the measurement, i.e. the experienced situation of the embedded observer (Doc. 270, Doc. 248), not by the geometry.

.3 “6 is the doubling of the 3” – literal (dim_bridge_1)

The icosahedral 6D representation used in the cut-and-project decomposes as

$$\mathbf{6} = \mathbf{3} \oplus \mathbf{3}', \quad (2)$$

the two inequivalent 3D irreducible representations of A_5 (physical $\oplus$ perp). On the 5-fold conjugacy class the characters are

$$\chi_{\mathbf{3}}(g_5) = 1 + 2 \cos 72^\circ = \varphi, \quad \chi_{\mathbf{3}'}(g_5) = 1 + 2 \cos 144^\circ = 1 - \varphi = -\frac{1}{\varphi}, \quad \chi_{\mathbf{6}}(g_5) = 1. \quad (3)$$

So the doubling is literal – $\mathbf{6}$ is two copies of a 3D space – and the golden ratio sits *in the split itself*: the two 3D blocks differ exactly by $\varphi \leftrightarrow 1 - \varphi$. The 6D character is the integer 1 while each 3D block is irrational (φ); this is the first half of why a 6D lattice exists where a 3D one cannot (next section). This φ in the doubling is HLV’s 5-fold / $2/\sqrt{5}$ at its root (cf. Doc. 281, Doc. 283).

.4 The triangle is fundamental, the pentagon needs the six (dim_bridge_2)

A rotation of order n is a lattice symmetry in 2D/3D if and only if $2 \cos(2\pi/n) \in \mathbb{Z}$ (crystallographic restriction). The integer values occur only for $n \in \{1, 2, 3, 4, 6\}$:

$$2 \cos \frac{2\pi}{3} = -1 \in \mathbb{Z} \quad (3\text{-fold, crystallographic}), \quad 2 \cos \frac{2\pi}{5} = \frac{1}{\varphi} \notin \mathbb{Z} \quad (5\text{-fold, forbidden}). \quad (4)$$

Hence 3-fold symmetry ($\mathbb{Z}_3$, Koide) lives natively in 3D – three dimensions *suffice*, and the triangle is the minimal, more fundamental carrier. 5-fold symmetry cannot sit in a 3D lattice; it exists only as a projection from 6D, where (by the previous section) it acquires an integer character. The doubling $3 \rightarrow 6$ is therefore not optional decoration: it is exactly what hosting a non-crystallographic symmetry costs.

.5 Same ambient $3+1$: what the cut-and-project family does and does not show (dim_bridge_3)

The cut-and-project ratio

$$r(\tau) = \frac{2\tau}{1 + \tau^2} \leq 1, \quad r(1) = 1, \quad (5)$$

is the family of cut-and-project “mathematical transformations.” HLV’s protected invariant $2/\sqrt{5} \approx 0.894$ is reached at $\tau = \varphi$ (and $\tau = 1/\varphi$). FFGFT’s $\sqrt{2} \approx 1.414$ lies *above the maximum of the entire family* and is reached by no τ whatsoever.

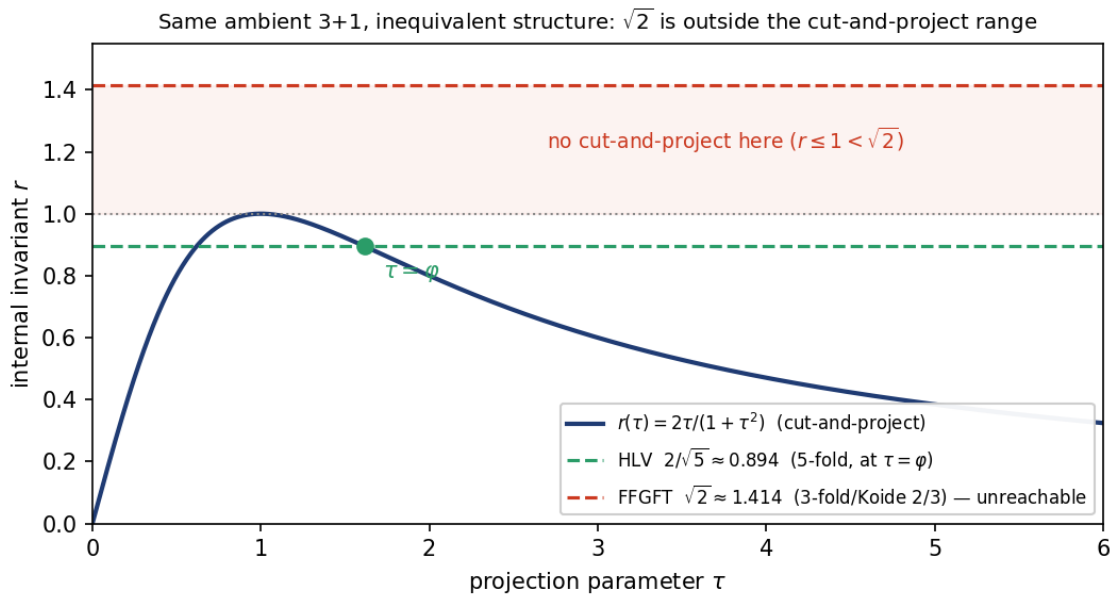


Figure 1: The cut-and-project invariant $r(\tau) \leq 1$ reaches HLV’s $2/\sqrt{5}$ at $\tau = \varphi$ but never FFGFT’s $\sqrt{2}$. That $\sqrt{2}$ lies above this family’s maximum means only that this particular projection does not interconvert the two invariants – not that the structures are incompatible (see next section).

What this shows – and what it does not:

- **Ambient $3+1$:** both models end in $\mathbb{R}^3 \times \mathbb{R}_t$; at the level of the bare coordinate manifold this is a lossless representational transformation (Type III, Doc. 270/230).
- **The cut-and-project map:** $\sqrt{2}$ lies outside the range $r \leq 1$, so *this* one-parameter family does not carry HLV’s $2/\sqrt{5}$ into FFGFT’s $\sqrt{2}$. That is a statement about one map, *not* a proof of incompatibility: reconciliation is not “interconvert the two invariants” but “embed both in a common structure” – the subject of the next section. (And $\sqrt{2} > 1$ versus $r \leq 1$ are not strictly commensurable, so their non-equality here is weak evidence at best.)

.6 The reconciliation: HL_V contains FFGFT (recompute_reconcile)

The decisive fact is group-theoretic. The icosahedral group A_5 *contains* the 3-fold C_3 : a 120° rotation about a $(1, 1, 1)$ axis and a 72° rotation about a vertex axis are *both* symmetries of the same regular icosahedron – both permute its twelve vertices (recompute_reconcile.py). Hence $C_3 < A_5$, and at the level of dimensions $T^7 = T^4 \times T^3$, i.e.

$$7 = \underbrace{3_{\text{phys}} + 1_{\text{time}}}_{\text{FFGFT } T^4} + \underbrace{3_{\text{perp}}}_{\text{HLV extension}} . \quad (6)$$

Concretely $T^4 \hookrightarrow T^7$, $(x, t) \mapsto (x, 0, t)$ with $0 \in T^3_{\text{perp}}$ – a totally geodesic submanifold: every FFGFT configuration is the special HL_V configuration with vanishing perp coordinate. The relation between the models is therefore *containment*, $\text{HLV} \supset \text{FFGFT}$: FFGFT is the minimal crystallographic 3-fold core, HL_V the icosahedral extension adding the three perp circles and the full 5-fold. The protected values $\sqrt{2}$ and $2/\sqrt{5}$ are invariants of the coexisting C_3 and A_5 sectors of the *same* containing structure; they need not, and do not, transform into one another – and that is no contradiction. The containment is at the level of the groups and dimensions; the *protected* invariant remains a separate, realised choice – FFGFT protects an independent 3-fold geometry ($\sqrt{2}$), HL_V the 5-fold ($2/\sqrt{5}$), and (Doc. 283) HL_V's protected vacuum directions do not include $\sqrt{2}$. Containment makes the models compatible without making them identical.

Where this is asserted. All of this lives in the compactified picture — the tori with every circle compact, where no circle is singled out as time. The bridge to physical, decompactified observables runs through the single step $S^1 \rightarrow \mathbb{R}$, the measurement (Doc. 270); the containment is therefore asserted upstream of the measurement, in the compact state, not on the physical-time axis.

.7 Resolution

The tension – “both reduce to 3+1, so the 3 and the 6 can only describe the same thing” – is resolved not by a fork but by *containment*. The 3 and the 6 share the same physical 3+1 core, and the models are mathematically reconcilable: $\text{HLV} \supset \text{FFGFT}$ in both the symmetry ($A_5 \supset C_3$) and the dimensions ($7 = 4 + 3$). What remains is not a mathematical incompatibility but the question of *which symmetry sector is physically realised*. FFGFT realises the minimal, crystallographic 3-fold core, and its invariant – Koide $\theta = 2/9$ – is empirically confirmed; HL_V adds the 5-fold/perp structure as an extension whose distinct observable (the icosahedral / quasicrystalline coherence signature, Doc. 282/283) is exactly what a test of HL_V against FFGFT would probe. The earlier wording is corrected in both directions: “only a mathematical transformation” is too weak (the realised sector is a genuine, empirically decidable feature) and “inequivalent, irreconcilable structures” is too strong (the embedding $\text{HLV} \supset \text{FFGFT}$ exists). The precise statement: a shared 3+1 core, with HL_V the icosahedral *extension* of FFGFT's minimal 3-fold core, the difference residing in the realised symmetry sector.

.8 Recursion in the unified compact picture: λ^n

The notation (T^N, G) extends to the *recursion*: in both models the self-similar structure is a dilation by a single eigenvalue λ , the recursion being λ^n . The recursion eigenvalue is *distinct* from the protected invariant ($\sqrt{2}$, $2/\sqrt{5}$); conflating the two is an error.

FFGFT: $\lambda = \xi = 4/30000$ (**Doc. 270**). The recursion ξ^n is one-sided contracting ($\xi, \xi^2, \dots$ vanish at once), giving the fractal correction

$$D_f = 3 - \xi = 2.999867, \quad (7)$$

a dimension just below 3 – near-crystalline. The protected invariant $\sqrt{2}$ (Koide 2/3) is a *separate* quantity, not the recursion scale.

HLV: $\lambda = \varphi$ (**golden inflation**). The icosahedral quasicrystal is self-similar under the golden inflation, with the Fibonacci recursion $\varphi^2 = \varphi + 1$. The Fibonacci substitution matrix (rows (1, 1), (1, 0)) has eigenvalues φ (expanding) and $-1/\varphi$ (contracting) – the *same* $\varphi \leftrightarrow -1/\varphi$ pair as the $\mathbf{3} \oplus \mathbf{3}'$ physical/perp split. Being two-sided, the recursion is self-similar at *all* scales: a genuine quasicrystal. The protected invariant $2/\sqrt{5}$ is again separate.

	recursion λ	character	protected inv.
FFGFT	$\xi = 4/30000$	one-sided contracting, $D_f = 3 - \xi$	$\sqrt{2}$ (Koide 2/3)
HLV	φ	two-sided ($\varphi, -1/\varphi$), $\varphi^2 = \varphi + 1$	$2/\sqrt{5}$

So the fork appears once more, now in the recursion: ξ tiny (one-sided, near-crystalline) versus φ golden (two-sided inflation). The extra contracting mode ($-1/\varphi$, the perp direction) is exactly HLV's extension over FFGFT, consistent with $\text{HLV} \supset \text{FFGFT}$. In each model one number governs both the protected geometry and the recursion.

A recursion *phase* (an analogue of FFGFT's Koide angle $\theta = 2/9$) is *not* fixed here. It would have to be *derived* from the A_5 representation; the naive $2/9 + 1/5 = 19/45$ is numerology, has no derivation, and is not adopted. Computed: `recursion_unify.py`.

Status of φ . The golden ratio acquires a meaning here that it lacked in pure FFGFT – but *not* as an FFGFT constant: as the derived eigenvalue of HLV's 5-fold / perp sector, hence a marker of the boundary $\text{HLV} \setminus \text{FFGFT}$. This is consistent with Doc. 190 (P35): there φ^N was the cautionary example of an arbitrary fitting base with no forward content; here φ is the eigenvalue of a named symmetry (golden inflation $\varphi^2 = \varphi + 1$), not a measured scale forced into an exponent. So φ does not become an FFGFT prediction.

.9 The HLV phase from A_5 : $\theta = 0$, the amplitude carries the difference

The protected geometry can be sharpened to a *phase*. FFGFT's Koide angle $\theta = 2/9$ is a phase offset in the 3-component circulant

$$\sqrt{m_k} = 1 + A \cos\left(\frac{2\pi k}{3} + \theta\right), \quad k = 0, 1, 2, \quad (8)$$

whose Koide ratio is $Q = (2 + A^2)/6$, *independent of θ* . The model content then splits cleanly into amplitude A and phase θ .

FFGFT. $A = \sqrt{2}$ (the 3-fold protected amplitude) and $\theta = 2/9$ – a *free*, empirical offset ($Q = 2/3$ for any θ ; $2/9$ is the measured Koide phase), giving $Q = 2/3$.

HLV. $A = 2/\sqrt{5}$ (the 5-fold/icosahedral amplitude: the 5-fold eigenvalue phases $e^{\pm 2\pi i/5}$ fix it through the inner product $1/\sqrt{5}$). The protected vacuum directions are symmetry-rigid, so there is *no free offset*: $\theta_{\text{HLV}} = 0$, giving $Q = (2 + \frac{4}{5})/6 = 7/15$ – exactly the value in the Doc. 283 fork table.

So A_5 's eigenvalue phases play explicit, distinct roles: the 3-fold phases $\pm 2\pi/3$ set the circulant *lattice* (both models); the 5-fold phases $\pm 2\pi/5$ (and $\pm 4\pi/5$ in $\mathbf{3}'$) set HLV's *amplitude*

$2/\sqrt{5}$ – not a free phase. The honestly derived HLV phase is therefore $\theta_{\text{HLV}} = 0$, and the whole model difference lives in the amplitude (protected invariant $\sqrt{2}$ vs $2/\sqrt{5}$, i.e. Koide $2/3$ vs $7/15$).

	amplitude A	phase θ	Koide $(2+A^2)/6$	source
FFGFT	$\sqrt{2}$	$2/9$ (free, empirical)	$2/3$	3-fold
HLV	$2/\sqrt{5}$	0 (rigid, protected)	$7/15$	5-fold

This also disposes of the naive guess $\theta = 2/9 + 1/5 = 19/45 \approx 0.422$: it matches *neither* the derived phase (0) *nor* the Koide ratio ($7/15 \approx 0.467$), and is adopted nowhere. The result is derived under the 3-component Koide parametrisation; $\theta_{\text{HLV}} = 0$ follows from the rigidity of the icosahedrally protected directions (cf. Doc. 283). Computed: `phase_from_a5.py`.

.10 How FFGFT unrolls: decompactification as the bridge to observation

FFGFT unrolls by letting **one circle of the torus T^4 become time through the measurement act** (type-I decompactification). The three remaining circles supply observed space but stay structurally untouched. The recursion ξ^n thereby becomes a continuous dilation flow with log-period $|\ln \xi| = \ln 7500 \approx 8.92$. Crucially, the unrolling is *not* part of the theory but the bridge to observation (Doc. 248, Doc. 270) — the torus remains the original, the unrolled space is the image.

The compact form

In the compact picture FFGFT is the four-torus $\mathcal{M} = T^4 = (S^1)^4$ with $G = \mathbb{Z}_3$. All four circles are *equivalent* — there is no distinguished time. The three space circles carry the gauge structure (connection Laplacian); the fourth is the mass circle with the duality $\tilde{T} \cdot m = 1$.

The measurement act unrolls one circle (type I)

Decompactification is not a mathematical limit $S^1 \rightarrow \mathbb{R}$ but a physical act: the measurement selects one circle as time and unrolls it. This is a property of the *embedded observer* (Doc. 248), not of the geometry. The type-I map $S^1_{\text{mass}} \rightarrow \mathbb{R}_t$ turns the dual mass $m = 1/\tilde{T}$ into rest energy; in doing so the winding number is discarded — the information-discarding step that forces the observer into time.

Type	What happens	Example
I	one circle becomes time (unrolled, winding discarded)	$S^1_m \rightarrow \mathbb{R}_t$
II	one circle stays compact (discrete spectrum)	$S^1_x \rightarrow S^1_x$ (unchanged)
III	lossless coordinate transformation	$T^4 \rightarrow \mathbb{R}^3 \times S^1$

Important for clean bookkeeping: *observed* space $\mathbb{R}^3$ arises from the lossless coordinate transformation (type III) $T^4 \rightarrow \mathbb{R}^3 \times S^1$ and carries continuous momenta; the *discrete, mass-bearing* spectrum, by contrast, sits on the compact $\mathbb{Z}_3$ fiber, not on the unrolled axis. "Compact/discrete" (fiber, masses) and " $\mathbb{R}^3$ /continuous" (type-III projection, space) are thus two different things and must not be conflated.

The recursion unrolls along with it

In the compact picture the recursion is discrete, λ^n with $\lambda = \xi = 4/30000 = 1/7500$. Upon unrolling this becomes a continuous dilation flow $t \mapsto \xi t$, $t = t_0 \xi^s$ ($s \in \mathbb{R}$) — the scale flow. A discrete scale invariance in the continuum is **log-periodicity** with period $|\ln \xi| = \ln 7500 \approx 8.92$ e-folds. This is a very long period; hence FFGFT is *near-crystalline*, the fractal correction tiny: $D_f = 3 - \xi = 2.9998667$. The detailed treatment (dilation flow, log-period, branching) follows in the next section.

What survives — and what does not

	Compact (T^4)	Decompactified ($\mathbb{R}^3 \times \mathbb{R}_t$)
Mass circle	S^1 , $m = 1/\tilde{T}$	$\mathbb{R}_t$, rest energy $E = m$
Space	$3 \times S^1$	$\mathbb{R}^3$ (type III, continuous momenta)
Mass spectrum	eigenvalues of S on the $\mathbb{Z}_3$ fiber (discrete)	unchanged; $m_k = \mu_k^2 M$, $M = 1/T$
Recursion	ξ^n (discrete)	ξ^s (flow), log-period $\ln 7500$
Symmetry	$\mathbb{Z}_3$	$\mathbb{Z}_3$ (preserved)
Observer	none	embedded (measurement)

The physics sits in the *compact* picture and survives the unrolling because it consists of *ratios*: the $\mathbb{Z}_3$ symmetry (on the fiber, not on the torus), the Koide structure ($r = \sqrt{2}$, $\theta = 2/9$, frame-independent), and the mass ratios. The mass operator $S = \text{circ}(t_0, t_1, t_2)$ does not change upon unrolling — it lives on the C_3 fiber, which is independent of the torus; the scale $M = 1/T$ comes from the time circle. Only the winding number of the unrolled circle is discarded. (Three time aspects: connection Laplacian without time, process time via the non-Markovian kernel, geometry time via $\tilde{T} \cdot m = 1$; Doc. 282.)

Consequence for the D7 comparison: FFGFT unrolls the mass circle into time and keeps the mass-bearing 3-core; Krüger's decompactified variant covers only $\{3' + \text{time}\}$ of this (below). Computed: `decompact_recursion.py`.

.11 Decompactified recursive time: continuous dilation flow and log-periodicity

Compact and decompact are related by $S^1 \rightarrow \mathbb{R}$ — the measurement (Doc. 270), and the recursion follows that map. In the compact picture the recursion is the *discrete* dilation λ^n ($n \in \mathbb{Z}$): a *discrete scale invariance* (invariance under scaling by λ only). When the mass circle unrolls to non-compact time, the discrete recursion becomes a *continuous* dilation flow

$$t \mapsto \lambda t, \quad t = t_0 \lambda^s, \quad s \in \mathbb{R}, \quad (9)$$

i.e. the renormalisation-group / scale flow. A discrete scale invariance in the continuum is *log-periodicity* of period $|\ln \lambda|$ — equivalently, complex scaling exponents $\alpha = \alpha_0 + i k 2\pi / |\ln \lambda|$.

	compact	decompactified
FFGFT	ξ^n (discrete)	flow $t \rightarrow \xi t$; log-period $ \ln \xi = \ln 7500 \approx 8.92$
HLV	φ^n (discrete)	flow $t \rightarrow \varphi t$; log-period $\ln \varphi \approx 0.481$

The fork persists in decompactified time. **FFGFT**: a log-period of ≈ 8.9 e-folds – very long, so barely log-periodic; near-crystalline. It is the same discrete scale invariance that gives $D_f = 3 - \xi$. **HLV**: a log-period of ≈ 0.48 e-folds – short and dense; a full quasicrystal, self-similar at every scale.

FFGFT's *recursion* R is precisely this object: discrete (ξ^n) in the compact picture, a continuous RG / scale flow once decompactified – the R reads both ways (recursion $\leftrightarrow$ renormalisation flow). Computed: `decompact_recursion.py`.

Marcel's decompactified variant: the overdamped kernel is the decompactified empty window

The doubling has two branches; decompactifying $S^1 \rightarrow \mathbb{R}$ (the measurement) turns each into a continuous scaling exponent $\alpha = \ln \mu$:

branch	$\text{Re } \alpha$	decompactified character
expanding φ (the D4 core, masses)	+0.481	relevant; coherence / revival (back-flow 5.125)
contracting $-1/\varphi$ (the $3'$ window)	-0.481	irrelevant; decaying / overdamped

Both share $|\ln \varphi| \approx 0.481$; only the sign differs. The decaying branch is exactly HLV's overdamped, decaying kernel (Docs. 282–284): **Marcel's overdamped kernel is the decompactified empty window** — the compact “empty $3'$ ” ($\rightarrow 0$ discretely) and the decompactified “overdamped, decaying” ($\text{Re } \alpha < 0$, $\rightarrow 0$ continuously) are the same object on the two sides of the measurement.

Crucially, the containment survives decompactification. Decomcompactified D7 is *not* the overdamped kernel alone: it is the D4 core (the relevant, coherence-carrying φ -branch with the masses and revival 5.125) *plus* the $3'$ window (the irrelevant overdamped branch). The D4 core remains contained inside decompactified D7 as its relevant sector; Marcel's kernel is the *extra* window, not the whole. This is just the decompactified image of $T^4 \subset T^7$.

Two caveats. The identification “overdamped kernel = decompactified window” is a proposed reading (both are the decaying, non-discriminating sector), not a theorem. And because decompactification is the measurement (information-discarding), this branch sits on the null-reproducible side by construction: the discriminating content (revival 5.125) is in the relevant D4 core, not in the overdamped kernel. Computed: `decompact_branches.py`.

Relation to Krüger's HLV kernel and claim-state note

Krüger's actual HLV memory kernel is a positive mixture of decaying exponentials, $K(t) = \sum_i w_i \tau_i^{-1} e^{-t/\tau_i}$ (Krüger, *Pre-Registered Null-Model Benchmark for HLV-Constrained Memory Kernels*, 2026). Every mode is pure decay (scaling exponent $\text{Re} < 0$), so the kernel is the decompactified contracting $-1/\varphi$ window of the previous subsection. Decisively, Krüger's own pre-registered benchmark finds this kernel statistically indistinguishable from a generic positive two-exponential kernel — the uniqueness gate stays false, “not a unique physical signature.” That is the null-reproducible side, established by his result, not merely asserted by ours.

The full HLV operator adds a skew-adjoint phase channel ($J^\top = -J$) to the positive kernel. The skew-adjoint (non-gradient) part is exactly the structure his claim-state note requires for sustained recurrence — pure gradient flow admits no cycles — and it is the analogue of the relevant, oscillatory FFGFT revival (5.125); the decaying kernel is the overdamped window. The two sectors line up: discriminating content sits in the recurrent/coherent (non-gradient) part, not in the decaying magnitude.

A third, sharper correspondence comes from the path-memory comparator (De Jesus, HICE-MEM, for Krüger): temporal order is antisymmetric, $\text{Area}(P) = \sum_{i < j} [A_i, A_j]$ with $\text{Area}(P^{\text{rev}}) = -\text{Area}(P)$, so a single history's magnitude and spectrum are *even* under reversal — the anti-Hermitian spectrum $\{i\lambda\} \mapsto \{-i\lambda\}$ is the same set. The order-sign is discarded unless contracted against an oriented reference (a second history). This is the same structure as the measurement $S^1 \rightarrow \mathbb{R}$ discarding winding while the magnitude survives: the discriminating content is relational, not single-history.

Finally the claim-state. Krüger's clarification note (2026) separates compact internal Spiral-Time, SI measurement time, and the signal index $\Delta\Phi_{\text{sig}}$, and forbids their identification without an explicit projection/calibration map K — the same discipline as "structure $\neq$ physics without a map." No identification of the two spiral notions and no holonomy is claimed here: Krüger's Spiral-Time enters as a bounded kinetic prefactor $A_\psi(t)$ on the kernel (benchmark) and as a triadic compact coordinate $\theta \in S^1$ (note), neither of which is the log-periodic log-spiral of the decompactified recursion. Computed: `marcel_kernel_match.py`.

Krüger's unrolled variant covers only $\{3' + \text{time}\}$ — not the full D7

A placement of the decompactified side, *without* importing Krüger's formulas. The compact FFGFT result — $6 = 3 \oplus 3'$ as a fractal internal recursion, purely from the golden ratio (Fibonacci operator, eigenvalues φ expanding / $-1/\varphi$ contracting) — is derived and script-covered (`recursion_doubling_empty.py`, `verify_recursion_doubling.py`, 11/11) and stands independently of HLV. It is not touched here; the comparison below does not bear on it.

Krüger's latest decompactified construction (*Recursive Spiral-Time Nesting as a Route to a Six-Dimensional Host Carrier*, 2026, with the null-gated runs Zenodo 20770402 / 20772007) reaches a 6D host via a *stability selection* (a spectral gap), *not* via φ — a different mechanism, no φ substitute. What matters for compatibility, however, is not the mechanism but *what the unrolled variant fills*: exclusively the **empty** sector. Its decaying kernel is the decompactified $-1/\varphi$ window (above); its host sector is candidate-only and massless ("host dimension, not masses"); and its compact time satisfies $\chi(t) \neq \ln t$ (clarification note), so it is not the log-spiral of the FFGFT recursion.

The full-text version (2026) adds the downstream chain BD3-BD5 (Zenodo 20773678): a native finite carrier (BD3), a low-q Lorentz/EFT pre-screen (BD4) and semantic transfer (BD5). It only adds more structure and still reaches no physics — BD4-HARD itself flags the low-q behavior as *carrier-conditioned*, *not HLV-specific* (matched nulls also pass), and BD0/BD1 remain negative. This reinforces the $\{3' + \text{time}\}$ picture: the carrier is more elaborate but massless structure. (Krüger dates the nesting idea itself to the January-2026 HLV foundation, Zenodo 18255159.)

In D7 bookkeeping — $7 = [3\text{-core (crystallographic, masses)} + \text{time}] + 3'$ — his variant therefore unrolls only $3' + \text{time}$: the empty perp window plus the unrolled time axis ($S_m^1 \rightarrow \mathbb{R}_t$, shared). The mass-bearing crystallographic 3-core (φ/ξ , $\sqrt{2}$, Koide 2/3) is *not reached* by it and remains FFGFT's. Strictly, his HLV — as decompactified-realized — is *not* the full D7 space, but the subsector $\{3' + \text{time}\}$; the full D7 (mass core + time + $3'$) is realized by FFGFT.

Hence compatibility, briefly: his side $\subseteq$ FFGFT's $\{3' + \text{time}\}$. The formal containment $T^7 \supset T^4$ ($A_5 \supset C_3$) stands; it is only sharpened here as to *where* Krüger's unrolled construction actually sits. This is compatible *as long as he holds the candidate-only line* (his gates B0-B7); were he later to overreach ("6D host = physical spacetime with masses"), compatibility would have to be re-examined — at present he explicitly does not. Covered by `decompact_branches.py` and the emptiness theorem above; no new script is needed for his formulas.

.12 What the d7 sector computes: masses from ξ , spectra from φ

By analogy with ξ one asks what the d7 model (T^7 , A_5) computes. The honest answer is *not masses*. Masses are the output of the crystallographic 3-fold sector with ξ , which d7 already contains ($\text{HLV} \supset \text{FFGFT}$); d7 reproduces them *through* its FFGFT sub-part, not as new content. The *extra* d7 structure (5-fold, perp directions, φ) is exactly the part that does not carry masses. What it carries instead:

quantity	d7 / 5-fold output	status
protected amplitude	$2/\sqrt{5}$, Koide-analog $Q = 7/15$	derived number
phase	$\theta_{\text{HLV}} = 0$ (no free Koide angle)	derived
recursion scale	φ ; tower φ^n ; log-period $\ln \varphi \approx 0.481$	derived
spectrum	aperiodic quasicrystal gap spectrum, φ -scaling	discriminator (271/283)

Why the spectrum does not lead to masses here

FFGFT's masses are themselves spectral – eigenvalues of the T^4 connection-Laplacian, the Koide circulant levels – so the question is fair; the answer is about *which* spectrum. A mass is a sharp, isolated level, so reading masses off a spectrum requires a *discrete point spectrum*. The 3-fold is lattice-compatible ($2 \cos(2\pi/3) = -1 \in \mathbb{Z}$): it closes on a periodic torus into exactly three sharp eigenvalues $\{-1, -1, 2\}$ – the triple. The 5-fold is lattice-incompatible ($2 \cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$): no periodic lattice carries it, so its only carrier is an aperiodic quasicrystal, whose spectrum is *singular-continuous* (Cantor-like) – a dense set of gaps, self-similar under φ , with no isolated levels (a Fibonacci chain fragments into gaps at every scale, against the few bands of the periodic chain; `spectrum_to_mass.py`). There is no distinguished triple to read masses off. Hence “spectrum $\rightarrow$ mass” needs a point spectrum, i.e. a periodic / crystallographic carrier – and the only crystallographic part of d7 is its 3-fold core, i.e. FFGFT. The route always returns to FFGFT; the genuinely-d7 (aperiodic) spectrum cannot deliver masses. And a dense self-similar spectrum has a level near *any* number, so φ^N “fits” anything – the P35 numerology trap (Doc. 190, R44): the very richness of the quasicrystal spectrum is what makes reading a mass off it content-free, whereas the sparseness of the discrete torus spectrum is what makes an eigenvalue there a genuine prediction.

Information content: structure spectrum vs energy spectrum

The previous point must not be read as “the quasicrystal spectrum carries no information.” It does – a quasicrystal has *two* spectra, differing in type. The *structure* (diffraction) spectrum $S(q) = |\sum_n e^{iqx_n}|^2$ is *pure point*: sharp Bragg peaks at φ -module positions, here $\approx 21\times$ above the noise floor of a random null of identical composition (`quasicrystal_information.py`). That sharp, null-distinguishable order *is* the information – exactly HLV's substrate-distinguishability claim, which stands. The *energy* (Laplacian) spectrum is the singular-continuous (Cantor) one of the previous subsection; it is what lacks sharp levels and so yields no masses. The two are different objects: information lives in the relational long-range order (structure spectrum), not in individual energy levels (the energy states are critical / fractal, not clean nodes). The P35 caveat is therefore narrow: only fitting a *single target number* as φ^N is content-free; the structural signature is not.

Residual, kept honest. The place where the FFGFT analysis found an HLV signature *null-reproducible* is the *temporal* over-damped memory kernel (Doc. 282–284, biexponential, non-identifiable sector) – the time domain, a different object from the spatial structure spectrum. So: spatial structure informative (HLV's point holds); temporal over-damped kernel = the standing critique. The discriminating content lives in the structural / spectral signature (pure-point diffraction; periodic-vs-aperiodic peaks) – the discriminator of Doc. 271/272/283. FFGFT and HLV agree *where* the information sits (in the structure); the critique concerned a different (temporal) claim.

The one place d7 could touch masses (conjecture)

The $6 = 3 \oplus 3'$ split leaves the perp irrep $3'$ (character $1 - \varphi = -1/\varphi$) as a *second* triplet position; FFGFT realises the one 3 (charged leptons, Koide $2/3$), the $3'$ is empty. *If* a physical triple sat there, its prediction would be $Q = 7/15$ – a falsifiable target, not a derived particle mass. Computed: `what_d7_computes.py`.

.13 The electron mode in D7: the embedding exists, the mass-label is C_3 -frame

A natural consistency question: if a mode is read off as the electron in D4, it must have an image in D7, since $T^4 \subset T^7$. It does. The electron mode lifts to the D7 mode with the same winding on the four D4 circles and zero winding on the extra three; sitting entirely in the D4 sub-block, its Laplacian eigenvalue — the mass — is unchanged. The translation is the embedding itself.

This does not conflict with “masses require the D4 core.” The sharp label “electron” is a C_3 -protected (crystallographic) statement, not an A_5 one: the electron mode is an eigenvector of the 3-fold. The full D7 symmetry is A_5 , whose 5-fold elements do not commute with C_3 — they rotate the electron mode into a muon/tau mixture. Under a genuine icosahedral 5-fold rotation the electron direction keeps 0.762 of its weight and leaks 0.238 into the μ/τ directions (`electron_mode_in_d7.py`). So in the full A_5 frame “electron” is not a sharp eigenstate; to read it as the electron (a sharp mass) one selects the C_3 eigenbasis — the restriction to the crystallographic sub-block, i.e. “going back to D4.” The correspondence is the embedding; the mass read-out is the frame choice; the two are the same statement.

Two safeguards. The mixing is a change of basis, not a smearing: the 5-fold rotates the labelling frame and does not destroy the physical mode, which stays perfectly sharp in any C_3 frame. And the mode stays in the periodic “3” block (point spectrum) — it does not leak into the aperiodic $3'$ sector; the 5-fold only rotates $e \leftrightarrow \mu \leftrightarrow \tau$ within the physical “3”. Finally, which frame counts as the 3-fold is, once again, the measurement (Doc. 270/248): the mode exists frame-independently in D7, the sharp particle label is frame-dependent.

.14 The $3'$ is the first recursion: why recursion directions are empty in the compact picture

The split $6 = 3 \oplus 3'$ and the golden recursion are not two facts but one: the doubling *is* the recursion acting once. This section makes that explicit and draws the consequence for the empty $3'$.

The doubling is the recursion operator

The Fibonacci / inflation operator is

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \det M = -1, \quad \text{tr } M = 1, \quad \text{characteristic polynomial } \lambda^2 - \lambda - 1 = 0. \quad (10)$$

Its two eigenvalues are $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ (expanding, $|\varphi| > 1$) and $-1/\varphi = \frac{1-\sqrt{5}}{2} \approx -0.618$ (contracting, $|-1/\varphi| < 1$), and $\varphi^2 = \varphi + 1$. Crucially, the Galois conjugate that distinguishes the two 3-dimensional A_5 irreps, $\varphi \leftrightarrow 1 - \varphi$, is exactly the contracting eigenvalue: $1 - \varphi = -1/\varphi$. So the two eigendirections of M are the two irreps:

3 = the φ (expanding) eigenmode, $3'$ = the $-1/\varphi$ (contracting) eigenmode.

The appearance of $3'$ alongside 3 is therefore one application of the golden recursion — the doubling is the recursion's eigenstructure. (That M is the recursion operator is confirmed by $M^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, the Fibonacci numbers.)

The internal branch is the window — hence empty

In the cut-and-project reading of a quasicrystal, physical space is the expanding direction (φ) and internal space is the contracting direction ($-1/\varphi$). Internal space carries the acceptance *window*, not physical positions. Under iteration the internal component decays geometrically while the physical component grows:

n	physical (φ)	internal ($3'$) $\sim (1/\varphi)^n$
0	1.01	0.27
2	2.64	0.10
8	47.4	0.0058
20	—	1.2×10^{-9}

The decay rate per step is $1/\varphi^2 \approx 0.382$. So the $3'$ (internal branch) carries no physical positions and vanishes under the recursion: that is why it is empty.

The general principle: the seed carries, the recursion is empty

The crystallographic seed (the " 3 ", φ -branch) holds the masses; everything the golden recursion generates from it — the conjugate $3'$ and the φ^n tower — is empty of *new* mass content. "Empty" means no new sharp level / no new triple, not "does nothing": the recursion still carries structure (FFGFT's ξ^n gives the fractal correction $D_f = 3 - \xi$; HLV's φ^n gives the quasicrystal window). Several earlier results collapse into a single statement:

$3'$ empty = recursion directions empty = masses need the crystallographic core = the recursion sector has no point spectrum = the cut-and-project internal space carries no physical positions.

Consequence for the $3'$ candidate: why a dynamics is needed

This gives a *structural* reason for the gap noted earlier. Putting a physical triple into the $3'$ means putting mass content into the internal / window space — precisely what the compact recursion keeps empty. It cannot be supplied by the recursion alone; it requires a dynamics that fills the window (a Hamiltonian / interaction on the 7D torus). So the conditional $3'$ candidate (the neutrino sector) needs a dynamics not only for methodological caution but for this structural reason: the recursion leaves that branch empty by construction.

Check scripts

- `recursion_doubling_empty.py` — demonstration: M 's eigenstructure ($\varphi / -1/\varphi$), physical vs internal under iteration.
- `verify_recursion_doubling.py` — assertions (PASS/FAIL): eigenvalues, char. poly, $1 - \varphi = -1/\varphi$, M^n Fibonacci, internal branch $\rightarrow 0$ at rate $1/\varphi^2$. 11/11 PASS.

.15 The recursion tower $D(4+3n)$: height is empty, depth needs dynamics

From the previous section the doubling is one golden recursion. The construction iterates: each recursion adds one 3 (the contracting $-1/\varphi$ branch), giving a dimension tower $D(4+3n)$, $n = 0, 1, 2, \dots$

n	dim	
0	D4	seed (FFGFT, crystallographic kernel)
1	D7	one recursion (the first 3')
2	D10	two recursions
3	D13	
100	D304	formally unbounded

But the theorem of the previous section fixes what this means: each added 3 is an *empty* internal window (the contracting $-1/\varphi$ branch, which $\rightarrow 0$ under iteration). Hence 100 recursions are 100 empty windows, *not* 100 physical layers; the masses reside, at *every* level of the tower, only in the D4 seed. The tower is a geometric hierarchy (nested cut-and-project), not new testable content — and the theorem is what keeps it from becoming a number mill (one could otherwise assert arbitrarily many “levels”, all empty).

Two caveats. (i) “+3 per step” assumes every level repeats the icosahedral 3+3 pattern; a different non-crystallographic symmetry (e.g. 8-fold, silver ratio $1 + \sqrt{2}$) would give a different increment, so $D(4+3n)$ is the icosahedrally-repeated choice, not mandatory. (ii) Filling any window requires a dynamics — the same open gap at every level; the tower does not escape the gap, it multiplies it.

Constructive consequence. The productive next step is not height (D10, D100 — more empty windows) but depth: the dynamics that fills the *first* window. And one physically interesting first window already exists — the $3' \rightarrow$ neutrino sector (conditional, next section). That is where the lever is, not in counting up the tower. Computed: `recursion_tower.py`.

Two recursions — and what the empty windows are good for

A natural tension: “each +3 step is one more recursion depth, recursions are fractal deepening — but if they are all empty and 4D already carries everything, why should unrolling yield mass?” It resolves once two *distinct* recursions, easily conflated, are separated:

- **Scale recursion** $\xi_{n+1} = \xi_n(1 - 100\xi)$: refines the *same* four dimensions across scales (log-period $\ln 7500$) and carries $D_f = 3 - \xi$ resp. K_{frak} — it adds *no* dimension. This is the actual “fractal deepening”.
- **Dimensional recursion** $D(4+3n)$ (golden operator, $\varphi/-1/\varphi$): stacks empty $3'/3''$ windows; the inner-branch weight decays $(1/\varphi^2)^n \rightarrow 0$, no point spectrum.

The mass content upon decompactification comes from unrolling the *mass circle* (4th circle of D4, $\tilde{T} \cdot m = 1$, type I), which supplies the *single* scale $M = 1/T$; the dimensionless ratios

were already present compactly in 4D. The Kaluza-Klein intuition (unroll an extra dimension $\rightarrow$ mass tower) thus applies to the mass circle, *not* to the $3'/3''$ windows: an empty window unrolled yields empty perp space, no mass tower, as long as no *dynamics* (a Hamiltonian on the window) fills it.

What, then, is the recursion good for if it yields no new masses? Not for levels, but for *structure*: the scale recursion carries the fractal correction (D_f, K_{frak}), the dimensional recursion carries the cut-and-project *window* that shapes the aperiodic order and the *spectral fine structure* (splitting, inharmonicities) on top of the crystallographic base levels — not new fundamental tones but their fine structure. The $3'$ window is in this sense a *slot* provided by the geometry; whether it gains physical content is decided by a dynamics (the conditional neutrino candidate, next section), not by mere unrolling. Computed: `klaer_rekursion_masse.py`.

.16 Candidate for the $3'$: the neutrino sector (conditional)

What could sit in the empty $3'$? An *open* question — a conditional target, not a prediction. The discriminator is the Koide ratio $Q = \sum m / (\sum \sqrt{m})^2$; the conditional $3'$ target would be $Q = 7/15 \approx 0.467$. The measured triples:

triple	Q	
charged leptons e, μ, τ (the " $3''$ ")	0.667 (= $2/3$)	far above target
up quarks u, c, t	0.849	far above target
down quarks d, s, b	0.731	far above target

All measured charged triples have $Q \geq 0.67$ — none matches $7/15$; they are excluded as a $7/15$ carrier (robust, also under the mass uncertainties). One open candidate remains: the **neutrinos**. Their Q is not fixed, because the absolute mass scale is unknown; in a scan (normal ordering) $Q \approx 0.586$ for a hierarchical spectrum and falls towards $1/3$ as the scale grows. It crosses $7/15$ at a lightest mass $m_0 \approx 1.9$ meV ($\sum m_\nu \approx 61$ meV, within the cosmological bound ~ 120 meV). Structurally this is the natural partner: leptons split into a charged triple (the " $3''$ ") and a neutral one (neutrinos), and the second 3 irrep is the slot for the neutral triple.

Status (strict). This is a conditional target, *not* a prediction. Q varies *continuously* with m_0 , so " $7/15$ is hit" carries content only if $7/15$ is fixed independently by a dynamics (the missing piece from the section "Status of the D7 mass calculation") *and* beats a null. It is exactly Marcel Krüger's structure: carrier = neutrinos, observable = Koide Q , null control = against generic triples. That lifts the question from "guessing" to a "pre-registrable target" — no more. Computed: `which_triple_in_3prime.py`.

Interpretation (conjecture, not derived from the theorems). That the recursion sector carries no point spectrum is consistent with the peculiar character of the neutrino sector. It yields, however, no masses, no Δm^2 and no mixing angles — these require the still-missing dynamics. And the tower adds empty windows, not additional neutrino species.

Which energy type fills the window — and the open computational gap. Through $\tilde{T} \cdot m = 1$ FFGFT enforces *two* energy types (Doc. 255): *static* (rest mass = topologically frozen winding, the crystallographic core) and *kinetic* (free; for massless states the only form, $E = hf$) — "mass is frozen kinetic energy". The dynamics that fills the $3'$ window — the

temporal, holographic scale evolution (Doc. 263) — therefore supplies the *kinetic* type, not frozen rest mass: a filled window is photon-/neutrino-like (massless / near-massless), not a massive triple. This matches the neutrino as a “near-massless photon” (Doc. 047, $m_\nu \sim \frac{\xi^2}{2} m_e$) and time-as-energy $\tilde{T} = 1/\max(m, \omega)$ (Doc. 067/080): for $m = 0$, $\tilde{T} = 1/\omega$.

Open computational gap. What is missing is the *derived* rule to compute the kinetic part, via the duality, as a ξ -fractal fraction of the static part. The framework exists (Doc. 080: $\tilde{T} = 1/\max(E, \omega)$, electron = rest + γ -kinetic, photon = purely kinetic) and *one* instance exists (Doc. 047: $\xi^2/2$ suppression — but as an ansatz, not derived; Doc. 204 has a ξ correction to the kinetic energy). The general derivation “kinetic = ξ -fractal fraction of the static, via $\tilde{T} \cdot m = 1$ ” is precisely the still-missing dynamics — the next computational step, not a solved piece.

.17 Status of the D7 mass calculation: the gap and the next step

The HLV construction provides a consistent geometric embedding (T^7 , A_5 , $3 \oplus 3'$) and qualitative predictions (5-fold coherence, $2/\sqrt{5}$ rather than $\sqrt{2}$). An explicit mass formula on T^7 does not exist, however — symmetry alone does not fix the mass matrix. What is missing is a specified dynamics (Hamiltonian, interactions) on the 7D torus. Until that is given, the 5-fold predictions stay qualitative (“there should be signatures”) rather than quantitative (“at this value”). FFGFT therefore remains the only model with an explicit, tested mass formula. HLV is a structural conjecture that needs a dynamics — and that is the next construction site.

.18 Where the fractal corrections sit

The fractal corrections sit at two causally chained levels — and explicitly *not* in the ratios.

Root: the conformal fractal metric

The fundamental fractal quantity is $\xi = 4/30000$ with the recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (Doc. 133/275). Its metric manifestation is *conformal*, not diagonal:

$$g_{\mu\nu}^{(\text{frac})}(x) = \eta_{\mu\nu}(1 + \xi f(x)), \quad D_f = 3 - \xi \quad (\text{Doc. 051}),$$

with the ξ correction specifically on the time component, $ds^2 = (1 + \xi\lambda_0/E_\xi) dt^2 - dx^2$ (Doc. 025). In the winding/harmonic structure it appears as the damping exponent $\tau^{-(1+\xi)}$ (Doc. 159). *Not* as a static metric $\text{diag}(1, 1, 1, 1+\xi)$ — this form is unsupported; the mass circle is distinguished via $\tilde{T} \cdot m = 1$.

Level 1: local spacetime porosity

The first manifest correction is the local spacetime dimension

$$D_f^{\text{space}} = 3 - \xi \approx 2.99987 \quad (\text{Doc. 133}),$$

a tiny deviation from integrality ($\sim 0.003\%$, effectively 3). This is *not* the type-II projection (Doc. 270: $T^4 \rightarrow T^3 \rightarrow \mathbb{R}^3$, clean, for large R); the porosity is a separate quantum-gravitational effect.

Level 2: effective dimension and K_{frak}

Through the cumulative RG run (17 orders of magnitude, Planck $\rightarrow$ electroweak) the porosity becomes an *effective* dimension $D_f^{\text{eff}} \approx 2.973$, which Doc. 133 states is “not identical” to D_f^{space} . From it the correction for absolute values emerges:

$$K_{\text{frak}} = \left(\frac{D_f^{\text{eff}}}{3} \right)^{D_f^{\text{eff}}/2} \approx 0.9867 \approx 1 - 100\xi.$$

The causal chain:

$$\underbrace{3 - \xi}_{\text{porosity } \sim 0.003\%} \xrightarrow{\text{RG run, 17 OoM}} \underbrace{D_f^{\text{eff}} \approx 2.973}_{\text{effective}} \xrightarrow{\text{formula}} \underbrace{K_{\text{frak}} \approx 0.9867}_{\sim 1.3\%}.$$

K_{frak} is thus *not* an independent correction “at the SI step” but the RG-cumulated consequence of $3 - \xi$; it becomes visible only in the absolute/SI transition (Doc. 122).

Where they do *not* sit

Quantity	contains ξ / K_{frak} ?
$r = \sqrt{2}$, $\theta = 2/9$	no (pure $\mathbb{Z}_3$ structure)
μ_k (eigenvalues of S)	no
mass ratios, α as ratio	no (K_{frak} cancels)
Koide $Q = 2/3$	no (pure $\mathbb{Z}_3$ structure)
$D_f^{\text{space}} = 3 - \xi$, D_f^{eff}	yes (dimension)
absolute values (m in kg, G in SI)	yes (K_{frak})

The corrections affect only the dimensions (D_f^{space} , D_f^{eff}) and the absolute values (K_{frak}). **Ratios are ξ - and K_{frak} -free** — the decisive statement from Doc. 101/122.

.19 From geometry to SI: the conversion

This closing bridge is *purely conventional* — it adds no physics but translates the geometric quantities into human-made units. The only fundamental object is the dimensionless number $\xi = 4/30000$; $c, \hbar$ are conventions (SI-2019), G is measurement data, and mass ratios, Koide $2/3$ and α are dimensionless ratios (Doc. 101, Doc. 122). The decisive separation: *ratios* are unit- and K_{frak} -free; the fractal correction $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ appears *only* in the step from ratio to *absolute* (SI) value. The “circularity” of the conversion is thus metrological (SI-2019), not a physical problem (Doc. 101).

The compact quantities (purely geometric)

Quantity	Compact value	Meaning
r	$\sqrt{2}$	$\mathbb{Z}_3$ circulant amplitude
θ	$2/9$	$\mathbb{Z}_3$ phase
ξ	$4/30000$	fractal recursion (fundamental)
μ_k	$\{0.04035, 0.58021, 2.37944\}$	dimensionless eigenvalues of S
Q	$2/3$	Koide invariant

These are unit-free and K_{frak} -free.

The scale M from the time circle

The mass circle S^1 carries, via $\tilde{T} \cdot m = 1$, the *only* dimensionful quantity $M = 1/T$ — a mass in natural units. It is fixed from *one* measured mass (the muon):

$$M = \frac{m_{\mu}^{\text{meas}}}{\mu_{\mu}^2} = \frac{105.6583755}{0.58021^2} \approx 313.86 \text{ MeV}.$$

All other masses are then *predicted* — that is the test of the theory.

The masses (ratio level, K_{frak} -free)

Lepton	μ_k	$m_k = \mu_k^2 M$ (MeV)	measured (MeV)
Electron	0.04035	0.51100	0.51099895
Muon	0.58021	105.658 (anchor)	105.6583755
Tau	2.37944	1776.98	1776.86

The mass ratios are unit- and K_{frak} -free: $m_{\mu}/m_e = 206.77$, $Q = 2/3$. (Electron to five figures, tau as a $\sim 0.007\%$ prediction; the μ_k follow from $r = \sqrt{2}$, $\theta = 2/9$, fixed by Koide + (e, μ).)

The fine-structure constant α (ratio level)

$$\alpha = \xi \left(\frac{\sqrt{m_e m_{\mu}}}{\text{MeV}} \right)^2.$$

With $\sqrt{m_e m_{\mu}} \approx 7.348 \text{ MeV}$: $\alpha \approx 0.007199$, i.e. $1/\alpha \approx 138.9$; the exact bridge value $E_0 = 7.398 \text{ MeV}$ gives $1/\alpha = 137.04$ (the $\sim 1.3\%$ is the radiative-correction level). α too is unit- and K_{frak} -free; E_0 is a bridge constant from the mass ratios (Doc. 101).

The gravitational constant G (absolute level, with K_{frak})

G is the only quantity that needs a genuine SI conversion ($[G] = \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$). The structural derivation (Doc. 006/122/133) yields it as an absolute value in which the fractal correction appears *only here*:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \cdot C_{\text{conv}} \cdot K_{\text{frak}}, \quad C_{\text{conv}} = 7.783 \times 10^{-3} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \text{MeV},$$

$$G_{\text{SI}} \approx 6.6745 \times 10^{-11} \text{ m}^3 \text{kg}^{-1} \text{s}^{-2} \quad (K_{\text{frak}} = 0.986; \text{ with } K_{\text{frak}} = 1 - 100\xi = 0.9867 : 6.679 \times 10^{-11}).$$

(Corrected on 29 Sept. 2026: previously $G_{\text{SI}} = \frac{\xi^2}{4M} \cdot \frac{\hbar c}{(1 \text{ MeV})^2} \cdot K_{\text{frak}} \approx 6.674 \times 10^{-11}$ – dimensionally inconsistent ($\hbar c / \text{MeV}^2$ has units m/J), and with the scale $M = 313.86 \text{ MeV}$ instead of m_e it gives 1.72×10^{-11} ; now the corpus form with m_e and C_{conv} .) This is the *computed* gravitational constant (not an input); the scale anchoring via the RG amplification is given in Doc. 133/275, not in this conversion.

The general conversion rule

$$\text{quantity}_{\text{SI}} = \text{quantity}_{\text{geom}} \times \text{conversion factors} \times K_{\text{frak}}$$

with the conversion factors $\hbar, c$ and the scale M (convention, SI-2019). The separation in one line:

Level	Quantities	K_{frak}
Ratio	$m_{\mu}/m_e, m_{\tau}/m_{\mu}, Q, \alpha$	cancels out
Absolute (SI)	$m_e, m_{\mu}, m_{\tau}, G$	appears

References: Doc. 101 (circularity is metrological), Doc. 122 (ratio vs absolute, K_{frak}), Doc. 133/006 (derivation of G with E_0, K_{frak} , scale).

.20 BD12-XCORE: the FFGFT mass-core declaration

Krüger's open question — what makes a *legitimate* mass-core docking criterion rather than a post-hoc fit — FFGFT answers operationally, not by assertion: with a *frozen, pre-registered* declaration in matched-null format (his BD style). The rule has *no* free parameters: $r = \sqrt{2}$, $\theta = 2/9$, and the Z_3 /Foot-Koide spectrum $\mu_k = 1 + \sqrt{2} \cos(\theta + 2\pi k/3)$.

It produces: Koide $Q = \Sigma \mu^2 / (\Sigma \mu)^2 = 2/3$ exactly (forced by $r = \sqrt{2}$); the lepton ratios m_{μ}/m_e to 0.001%, m_{τ}/m_e to 0.007%; and — as a *genuine prediction* from Koide + $\{m_e, m_{\mu}\}$, with no τ input — $m_{\tau} = 1776.97$ MeV against 1776.86 measured (0.006%). The matched-null test (tolerance Koide 0.1%, ratio 1%) shows: random r hits Koide in 0.18%, random θ both ratios in 0.025%, random (r, θ) the *joint* set in 0.000% of cases. The frozen rule is therefore *special*, not a generic fit — **PASS**; and symmetric, random inputs would give **FAIL**.

Two constraints keep the dock honest: (i) do *not* dock on L_{nat} — the masses are not in the carrier Laplacian but in the compact Z_3 core + mass-circle ($\tilde{T} \cdot m = 1$); (ii) the dock is a *fixed transformation* (a pre-registered algebraic map onto $\{\sqrt{2}, 2/9, M = 1/T\}$), adding no new physics — hence *not* a post-hoc fit, but an equivalence that reproduces or fails. Computed: bd12_xcore_ffgft.py.

.21 Reproducibility

- dim_bridge_1_doubling.py – $\mathbf{6} = \mathbf{3} \oplus \mathbf{3}'$; characters $\varphi, 1 - \varphi$, integer 6D character; orthogonality and order 5.
- dim_bridge_2_crystallographic.py – crystallographic restriction $2 \cos(2\pi/n)$; 3-fold integer, 5-fold = $1/\varphi$.
- dim_bridge_3_same_ambient_fork.py – $r(\tau) \leq 1$; $2/\sqrt{5}$ at $\tau = \varphi$; $\sqrt{2}$ above the family's maximum (a statement about one map only).
- recompute_reconcile.py – $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron) and $7 = 4 + 3$; the containment $\text{HLV} \supset \text{FFGFT}$.
- recursion_unify.py – recursion λ^n : $\lambda = \xi$ ($D_f = 3 - \xi$) vs $\lambda = \varphi$ ($\varphi^2 = \varphi + 1$).
- phase_from_a5.py – the A_5 phase: $\theta_{\text{HLV}} = 0$, $Q = 7/15$; 19/45 excluded.
- decompact_recursion.py – decompactified recursion: discrete $\lambda^n \rightarrow$ continuous flow $t \rightarrow \lambda t$, log-period $|\ln \lambda|$.
- what_d7_computes.py, spectrum_to_mass.py – what the d7 sector computes (no masses; $Q = 7/15$, φ , spectra) and why a point spectrum is needed for masses.

- `quasicrystal_information.py` – structure (diffraction) spectrum pure-point vs random null: the substrate’s information content.
- `electron_mode_in_d7.py` – the electron mode embeds in D7 ($T^4 \subset T^7$); a 5-fold rotates it (0.762/0.238), so the sharp mass-label is C_3 -frame = the D4 core.
- `which_triple_in_3prime.py` – candidate triples for the 3’ via Koide Q : leptons/up/down excluded ($Q \geq 0.67$), neutrinos the open candidate (crosses 7/15 at $m_0 \approx 1.9$ meV).
- `recursion_doubling_empty.py`, `verify_recursion_doubling.py` – the 3’ as the contracting $(-1/\varphi)$ recursion eigenmode; internal branch $\rightarrow 0$ (11/11 PASS).
- `recursion_tower.py` – the tower $D(4 + 3n)$ (D4/D7/D10/.../D304) and the emptiness of every added window.
- `decompact_branches.py` – the two branches decompactified ($\text{Re } \alpha = \pm 0.481$): φ relevant (D4 core), $-1/\varphi$ overdamped (Marcel’s kernel); D4 core stays contained.
- `marcel_kernel_match.py` – grounded match to Krüger’s kernel $K(t) = \sum_i w_i \tau_i^{-1} e^{-t/\tau_i}$ (overdamped = $-1/\varphi$ window) and the antisymmetric path-memory order (single-history spectrum even under reversal).
- `bd12_xcore_ffgft.py` – the BD12-XCORE declaration: frozen rule ($r = \sqrt{2}$, $\theta = 2/9$), Koide $2/3$, ratios, τ -prediction and the matched-null test (nulls 0.000% on the joint set).

All scripts are numpy-only with fixed values. Note: the $\sqrt{2}$ appearing at $n = 8$ in the crystallographic table is $2 \cos 45^\circ$ (the octagonal rotation), a numerical coincidence – not FFGFT’s $\sqrt{2}$, which is the Koide/ $\mathbb{Z}_3$ -circulant ratio *within* the crystallographic 3-fold structure, not a rotation order.